

ST7072CEM Machine Learning

Coursework: Individual Report

Crop Recommendation from Soil and Climatic Parameters

A Comparative and Robustness Evaluation of KNN, Tree-Based and Naive Bayes Classifiers

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Abstract

Choosing a crop that suits local soil and climate is one of the most consequential decisions a farmer makes, yet it is often based on tradition rather than measurement. This report compares three families of classifiers—K-Nearest Neighbours (KNN), tree-based models (a single Decision Tree and a Random Forest) and Gaussian Naive Bayes—for recommending one of 22 crops from seven soil and climate features, using the public Crop Recommendation dataset (2,200 samples). PCA and K-means are used to explain the structure of the feature space. Under identical, leakage-free pre-processing and grid-search tuning, Random Forest and Gaussian Naive Bayes both reach 99.55% test accuracy and are statistically indistinguishable over 50 cross-validation folds (corrected resampled t-test, $p = 0.87$), while KNN (98.18%) and the single tree (97.95%) are significantly worse. Exploratory analysis suggests that the dataset is synthetic: within each crop, 141 of 154 feature distributions are consistent with uniformity and none with normality. Under simulated measurement noise KNN degrades most gracefully, while Random Forest leads with very scarce training data. Probability metrics disagree: Gaussian NB has the lowest log-loss, Brier score and top-label calibration error, yet its errors carry high confidence; Random Forest is less confident on its errors. A top-3 prototype is provided, with out-of-range input warnings. These results show that benchmark accuracy is close to saturated and that model choice depends on the deployment risk and the probability measure used.

Keywords: crop recommendation, precision agriculture, random forest, naive Bayes, k-nearest neighbours, robustness, calibration

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1 Introduction

Agriculture employs a large share of the workforce in low- and middle-income countries and remains central to food security and rural incomes [1]. In South Asia, including Nepal and India, most farms are small and crop decisions are commonly made from experience, local custom or neighbours' choices rather than from soil tests and climate records. A poorly matched crop can lower yields, waste fertiliser, deplete soil nutrients and leave the household exposed when rainfall or temperature departs from the norm. Precision agriculture aims to replace such guesswork with data-driven advice [2, 3]. A crop recommender is one of its simplest and most widely studied tools: given measurements of a field's soil nutrients and climate, it suggests the crops most likely to grow well there.

Machine learning (ML) suits this task because the link between growing conditions and suitable crops is non-linear and threshold-like. Rice, for example, needs high rainfall and humidity; chickpea tolerates dry conditions; apple and grapes need soils very rich in phosphorus and potassium. Rules like these are hard to write by hand for many crops at once, but a classifier can learn them from historical records.

1.1 Related work

Early crop-recommendation systems used classical classifiers on regional soil and weather data. Pudumalar et al. [5] proposed a recommender for Indian farmers that combines several base learners (including Naive Bayes and KNN) by majority voting. Kulkarni et al. [6] also used a majority-voting ensemble of heterogeneous classifiers and reported accuracy above 99%. Doshi et al. [7] built AgroConsultant, which pairs ML-based crop suitability prediction with weather and market information to give farmers more complete advice. Rajak et al. [8] applied an ensemble of classifiers to soil-test data to maximise yield. Jain and Ramesh [9] studied how weather variables can drive crop selection. Broader reviews [3, 4] show that crop management and yield prediction are among the most common ML applications in agriculture, and that tree ensembles, support vector machines and, more recently, deep networks dominate the field.

Three gaps recur in this literature. First, most studies report a *single headline accuracy* from one train/test split, without variance estimates or statistical tests. With many papers above 95%, differences of one or two percentage points are often within sampling noise [23, 25]. Second, algorithm families are rarely compared under *identical* pre-processing and tuning budgets, so it is unclear whether a reported advantage comes from the algorithm or from the experimental set-up. Third, little attention is paid to *why* models fail. This includes the structure of the feature space, the quality of the data, and model behaviour under the noisier conditions that real deployment brings. For a farmer-facing tool, a model that is 99.5% accurate on clean data but confidently wrong on slightly noisy inputs may be worse than one that is 98% accurate and honest about its uncertainty.

1.2 Contributions

This report addresses these gaps on the widely used public Crop Recommendation dataset [10]. Its contributions are:

- A controlled comparison of KNN, Decision Tree, Random Forest and Gaussian Naive Bayes with

leakage-free pipelines, cross-validated grid search, repeated cross-validation and two significance tests.

- An analysis of the feature space with PCA and K-means, linking unsupervised structure to the classifiers' confusion patterns. It includes evidence that the dataset is synthetically generated, which has not been reported alongside the many near-perfect accuracies published for it.
- Robustness and probability diagnostics—feature importance, feature ablation, measurement noise, data scarcity, and calibration—showing that clean-data rankings alone do not establish deployment suitability.
- A prototype recommender that returns the top three crops with probabilities and warns when inputs fall outside the training range.

The rest of the report is organised as follows. Section 2 describes the problem and the dataset. Section 3 introduces the methods and Section 4 the experimental set-up. Results appear in Section 5 and are discussed in Section 6, with social, ethical, legal and professional issues in Section 7. Section 8 concludes.

2 Problem and Dataset

2.1 Problem definition

The task is *multi-class classification*: given a vector $\mathbf{x} \in \mathbb{R}^7$ of soil and climate measurements for a field, predict the crop $y \in \{1, \ldots, 22\}$ best suited to those conditions. Every crop is equally important to the farmer who grows it, so performance is summarised with macro-averaged metrics. Because a recommender advises rather than decides, the *ranking* of crops and the *confidence* attached to it also matter, not only the top prediction. The study is guided by three research questions:

RQ1 How accurately can soil and climate measurements identify the most suitable of 22 crops?

RQ2 Which classification technique is best suited to the problem, and why?

RQ3 Do natural clusters in the feature space align with agronomic crop groups, and does PCA reveal a well-separated space?

2.2 Dataset

The Crop Recommendation dataset [10] contains 2,200 records, each labelled with one of 22 crops: rice, maize, chickpea, kidney beans, pigeon peas, moth beans, mung bean, black gram, lentil, pomegranate, banana, mango, grapes, watermelon, muskmelon, apple, orange, papaya, coconut, cotton, jute and coffee. Each record has seven numeric features: the soil's nitrogen (N), phosphorus (P) and potassium (K) content (kg/ha), temperature (°C), relative humidity (%), soil pH and rainfall (mm). Table 1 summarises the features.

Table 1: Summary statistics of the seven features ($n = 2{,}200$).

Feature	Mean	SD	Min	Median	Max
N (kg/ha)	50.6	36.9	0.0	37.0	140.0
P (kg/ha)	53.4	33.0	5.0	51.0	145.0
K (kg/ha)	48.2	50.7	5.0	32.0	205.0
Temperature (°C)	25.6	5.1	8.8	25.6	43.7
Humidity (%)	71.5	22.3	14.3	80.5	100.0
pH	6.5	0.8	3.5	6.4	9.9
Rainfall (mm)	103.5	55.0	20.2	94.9	298.6

2.3 Data quality and exploratory findings

The data has no missing values and no duplicate rows, and the classes are perfectly balanced at 100 samples per crop. Features differ greatly in scale (pH spans about 6 units, rainfall about 280 mm), which matters for distance-based methods. The only strong linear correlation is between P and K ($r = 0.74$), driven by apple and grapes, which require very high levels of both. All other pairwise correlations are below 0.25 in magnitude, so multicollinearity is not a concern.

The global interquartile-range (IQR) rule flags many points as outliers (e.g. 200 for K and 138 for P). Applied *within* each crop, however, it flags none at all. The global "outliers" are therefore the defining requirements of particular crops, not measurement errors, and no rows were removed.

Zero within-crop outliers across 2,200 rows is implausible for field measurements, so the shape of every crop's feature distributions was tested. For each of the $22 \times 7 = 154$ crop–feature pairs, a Kolmogorov–Smirnov test against a uniform distribution on the observed range and a Shapiro–Wilk normality test were applied. **141 of 154 pairs are consistent with a uniform distribution and none with a normal distribution** ($\alpha = 0.05$). The median excess kurtosis per feature lies between -1.13 and -1.22, matching the theoretical -1.2 of a uniform distribution. Rice, for instance, always has K between 35 and 45 and humidity between 80% and 85%, with values spread evenly in between (Fig. 1). The most plausible explanation is that the dataset was *generated* by sampling uniformly from each crop's recommended agronomic ranges, not collected from real farms. This finding shapes the rest of the study. It predicts very high accuracy, since classes are box-shaped regions that rarely overlap. It means Naive Bayes' Gaussian assumption is violated. And it motivates the robustness experiments in Section 5.4.

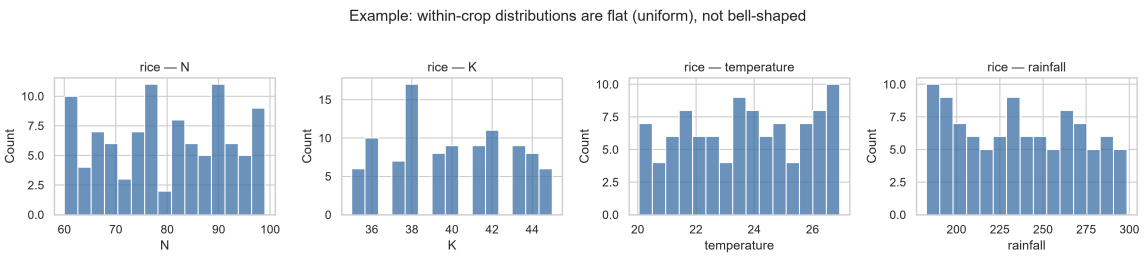


Figure 1: Within-crop distributions of four features for rice. Values are spread evenly between hard limits, characteristic of synthetic uniform sampling rather than field measurements.

3 Methods

The brief requires three classification or clustering techniques. Three classifier families were chosen to match different properties of the data: KNN (instance-based), tree-based models (rule-based) and Gaussian Naive Bayes (probabilistic). PCA and K-means were added as exploratory tools.

3.1 K-Nearest Neighbours

KNN [11] assigns a sample the majority label among its k nearest training samples, optionally weighting each neighbour by inverse distance. It makes no assumption about the shape of class boundaries, which suits a problem where crops with similar requirements sit close together. With only seven meaningful features, the curse of dimensionality is not a concern. KNN is sensitive to feature scale, so features are standardised to zero mean and unit variance before distances are computed. The tuned hyperparameters are k, the weighting scheme and the distance metric (Euclidean or Manhattan).

3.2 Decision Tree and Random Forest

A classification tree [12, 13] recursively splits the feature space with axis-aligned thresholds chosen to maximise node purity (Gini impurity or entropy). Agronomic suitability is naturally expressed as thresholds ("rainfall above 180 mm and humidity above 80%"), so trees match the domain closely and are easy to interpret. Single trees, however, have high variance. A Random Forest [14] averages many trees, each grown on a bootstrap sample and restricted to a random subset of features at every split. This decorrelates the trees and reduces variance. The single tree is reported as an interpretable baseline and the Random Forest as the main tree-based method. Trees are invariant to monotonic feature transformations, so no scaling is applied.

3.3 Gaussian Naive Bayes

Gaussian Naive Bayes (GNB) applies Bayes' theorem with the "naive" assumption that features are conditionally independent given the class, and models each feature within a class as a normal distribution:

$$\hat{y} = \arg\max_{c} \; P(c) \prod_{j=1}^{7} \mathcal{N}\left(x_j \mid \mu_{cj}, \sigma_{cj}^2\right). \tag{1}$$

It needs only $2 \times 7 \times 22 = 308$ parameters, trains in a single pass and gives class probabilities directly, which makes it a fast and transparent baseline. The `var_smoothing` hyperparameter adds a small constant to every variance for numerical stability. As Section 2 showed, the Gaussian assumption does not hold for this dataset. Naive Bayes is nonetheless known to classify well when its distributional assumptions are violated, because only the *ranking* of class posteriors must be correct, not their values [15–17].

3.4 Principal Component Analysis

PCA [18] finds orthogonal directions of maximal variance in the standardised feature space. Seven features are already few, so PCA is used here for *interpretation*: to visualise how separable the crops are, to identify which features drive the main directions of variation, and to test how much class information is lost when dimensions are discarded.

3.5 K-means clustering

K-means [19] partitions the data into k clusters by minimising within-cluster squared Euclidean distance. It is applied to the standardised features *without labels* in order to see whether the crops form natural groups. The number of clusters is examined with the elbow method and the silhouette coefficient [20]. Agreement between clusters and crops is measured with the Adjusted Rand Index (ARI) [21], normalised mutual information (NMI), homogeneity and completeness.

4 Experimental Setup

4.1 Pre-processing and data split

Crop names were label-encoded and the data was split into stratified training (80%, 1,760 samples) and test (20%, 440 samples) sets, giving exactly 80 and 20 samples per crop. The split used a fixed random seed (42), as did every other random process. The test set was held out from preprocessing and hyperparameter selection, and provides the primary model comparison. It is reused for post-hoc feature and robustness diagnostics in Section 5.4; these secondary analyses are descriptive and are not independent confirmatory estimates. Standardisation was placed *inside* a scikit-learn `Pipeline` for KNN and GNB, so during cross-validation the scaler is fitted on the training folds only. This avoids the subtle data leakage that arises when the whole dataset is scaled before splitting [22]. No feature selection was performed. Since the outliers were genuine crop requirements and no values were missing, no rows were altered.

4.2 Hyperparameter tuning

Each model was tuned by exhaustive grid search with stratified 5-fold cross-validation on the training set, optimising macro-F1. Table 2 lists the search spaces and selected values.

4.3 Evaluation protocol

The tuned models were refitted on the full training set and evaluated on the test set using accuracy, macro-averaged precision, recall and F1, per-class F1 and confusion matrices. Training time and per-sample inference time were also recorded.

A single 440-sample test set can only resolve differences in steps of 0.23% (one sample). To check whether the ranking is stable, each tuned configuration was therefore also evaluated with 10-times-

Table 2: Hyperparameter grids and selected values (5-fold CV on the training set, macro-F1).

Model	Grid	Selected
KNN	$k \in \{1,3,\ldots,25\}$; uniform/distance weights; Euclidean/Manhattan (52 combinations)	$k = 5$, distance, Manhattan
Decision Tree	depth $\in$ {None,5,10,15,20}; min. leaf $\in \{1,2,5\}$; Gini/entropy (30)	Gini, no depth limit, min. leaf 2
Random Forest	trees $\in \{100,300\}$; depth $\in$ {None,10,20}; min. leaf $\in \{1,2\}$; max. features $\in \{\sqrt{p}, p\}$ (24)	100 trees, depth 10, $\sqrt{p}$, min. leaf 1
Gaussian NB	var_smoothing $\in \{10^{-12},\ldots,10^{-2}\}$ (11)	10^{-12}

repeated stratified 5-fold cross-validation (50 fits per model). Pairs of models were compared with two tests: the corrected resampled t-test of Nadeau and Bengio [23] on the paired fold scores, which inflates the variance to account for overlapping training sets, and McNemar's exact test [24, 25] on the test-set predictions. Learning curves were computed with 5-fold CV over eight training-set sizes.

4.4 Robustness experiments

Four post-hoc stress tests probe behaviour beyond the clean benchmark, reusing tuned hyperparameters and the fixed test set. They are descriptive diagnostics, not independent validation:

1. *Permutation importance*: the drop in test accuracy when one feature's values are shuffled (20 repeats).
2. *Feature ablation*: each model retrained and tested without each feature in turn, simulating a missing measurement.
3. *Measurement noise*: Gaussian noise with standard deviation equal to 5–50% of each feature's training SD added to test inputs (20 repeats per level), simulating cheap sensors or approximate soil tests.
4. *Data scarcity*: models trained on 3–80 random samples per crop (10 subsets per size), simulating a new region with little labelled data.

In addition, predicted probabilities were assessed on clean and noisy (20%) test inputs with log-loss, the multiclass Brier score (mean sum of squared one-hot errors; range 0–2), top-label expected calibration error (ECE; ten equal-width confidence bins), mean confidence on errors, and the share of predictions above 0.99 confidence. A top-label reliability diagram compares mean predicted confidence with observed accuracy. All probability results are post-hoc diagnostics on this dataset and do not establish calibration under field deployment.

4.5 Implementation

All experiments were run in a Jupyter notebook with Python 3.12, scikit-learn 1.9 [27], pandas 3.1, NumPy 2.5, SciPy 1.18 and Matplotlib/Seaborn, on a Windows 11 laptop CPU. The full notebook executes in about three minutes. The code, figures, notebook and run instructions are available in the

project repository at `https://github.com/rajushah404/ST7072CEM-Machine-Learning`; a packaged copy accompanies this report.

5 Results

5.1 Classification performance

Table 3 reports test-set performance. Random Forest and Gaussian NB each misclassify only 2 of 440 test samples (99.55% accuracy, macro-F1 0.995). KNN makes 8 errors and the single Decision Tree 9. Precision and recall are nearly identical for every model, as expected for balanced classes. Nine of the 22 crops are classified perfectly by all four models.

Table 3: Test-set performance (440 samples) and 10×5-fold repeated CV accuracy (mean ± SD, 50 folds).

Model	Acc.	Macro-F1	Errors	Repeated CV acc.
KNN	0.9818	0.9817	8	0.9805 ± 0.0061
Decision Tree	0.9795	0.9794	9	0.9867 ± 0.0066
Random Forest	**0.9955**	**0.9955**	**2**	**0.9949** ± 0.0029
Gaussian NB	**0.9955**	0.9954	**2**	0.9946 ± **0.0022**

The errors are concentrated in a few groups of crops (Fig. 2). Rice misclassified as jute is the single error that *every* model makes. The other main group is the pulses: the Decision Tree confuses black gram, lentil and moth beans repeatedly, and its lowest per-class F1 scores are lentil (0.88) and black gram (0.89). Random Forest's only errors are one rice→jute and one black gram→maize, and Gaussian NB's are two rice→jute.

5.2 Stability and statistical significance

Repeated cross-validation confirms the ranking (Table 3, Fig. 3). Random Forest (99.49%) and Gaussian NB (99.46%) are statistically indistinguishable (corrected $t = 0.17$, $p = 0.87$; McNemar $p = 1.0$). Both are significantly better than KNN ($p < 0.001$) and the single tree ($p = 0.015$ and $p = 0.029$). KNN and the Decision Tree do not differ significantly ($p = 0.11$). On the test set alone, McNemar's test is significant for the tree comparisons ($p = 0.016$ and 0.039) but not for KNN vs. RF/GNB ($p = 0.07$). Only eight samples are discordant in those comparisons, which shows why repeated CV is the stronger evidence. Gaussian NB also has the smallest spread across folds.

The learning curves (Appendix, Fig. 12) show that Gaussian NB reaches its plateau with only about 300 training samples (about 14 per crop), and that Random Forest converges almost as quickly. KNN levels off near 97.5%, while the Decision Tree's validation accuracy is still rising at the full training size. In all cases the gap between training and validation accuracy narrows as data grows, so overfitting is limited.

Computationally, all models are cheap at this scale. In a representative run, KNN, the Decision Tree and GNB trained in about 10 ms or less and predicted in about 1–11 μs per sample. Random Forest took about 260 ms to train and about 0.1 ms per prediction. Timings varied by up to about 50% between runs.

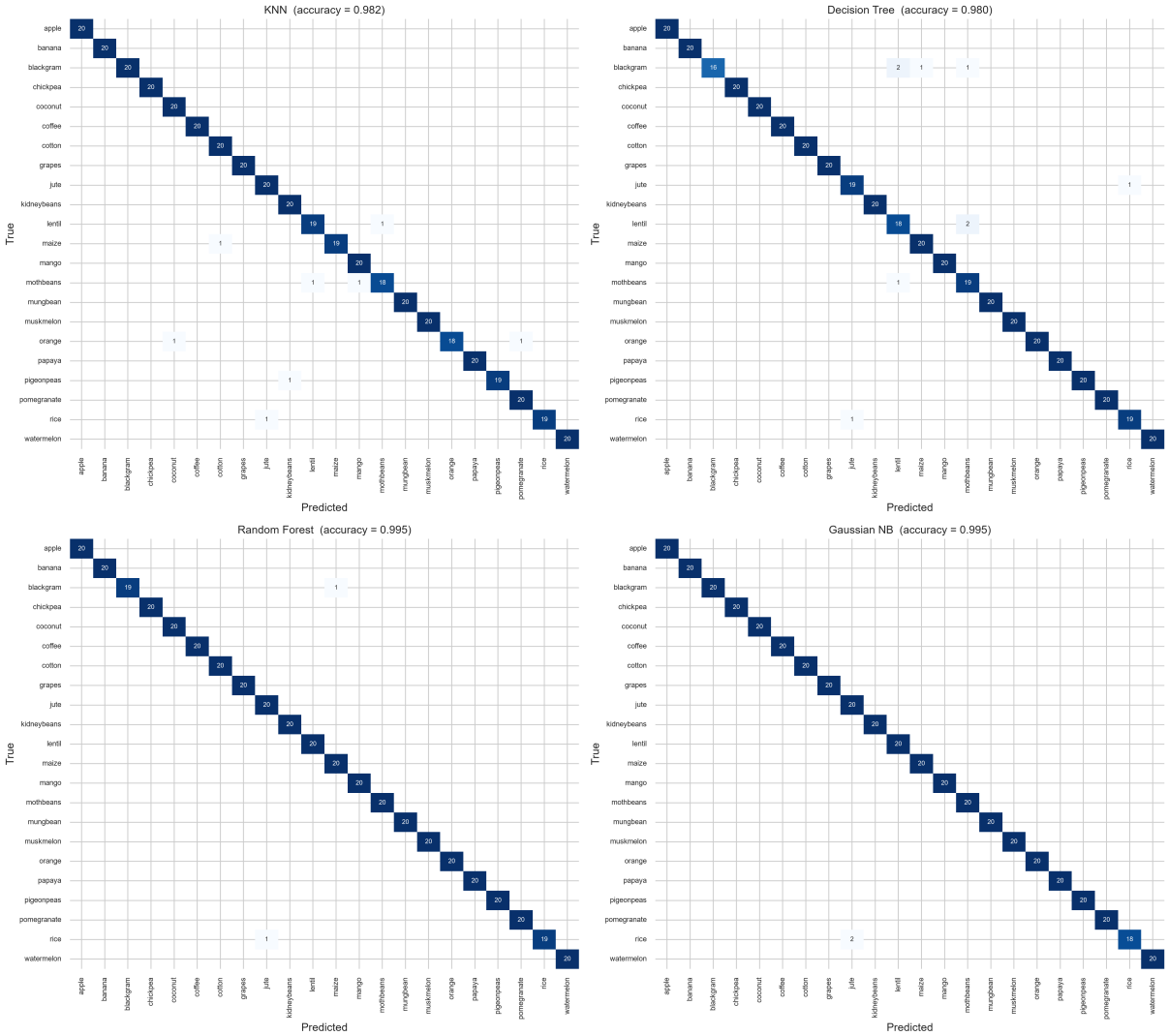


Figure 2: Test-set confusion matrices (zeros hidden). Errors concentrate in rice/jute and among the pulses (black gram, lentil, moth beans).

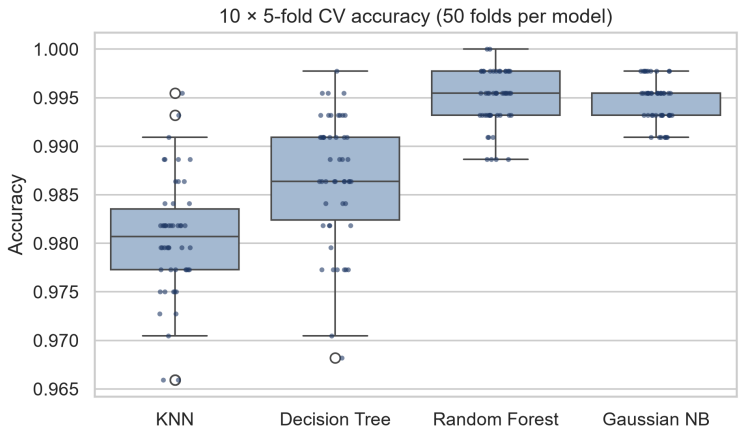


Figure 3: Accuracy over 50 folds of 10×5-fold repeated stratified CV.

5.3 Structure of the feature space

PCA. Variance is spread across components: the first two explain 27.6% and 18.5% (46.1% together), and six components are needed to pass 95%. PC1 is dominated by P and K (loadings 0.64 and 0.62) and separates apple and grapes from everything else. PC2 is driven by humidity (0.74), temperature and N, and separates the dry-climate pulses chickpea and kidney beans. In the 2-D projection (Fig. 4) these two groups are isolated, while the remaining 18 crops overlap heavily in the centre. Discarding dimensions destroys class information: KNN accuracy falls from 98.2% with all seven components to 94.3% with five and 46.8% with two. PCA is therefore useful for interpretation here but not for dimensionality reduction.



Figure 4: Projection onto the first two principal components (46.1% of variance). Only apple/grapes and chickpea/kidney beans separate clearly in two dimensions.

K-means. The silhouette coefficient is highest at $k = 2$ (0.42), reflecting a split between the high-P/K fruit crops and the rest. No pronounced elbow appears near $k = 22$, so the 22 crops do not emerge as natural clusters. With $k = 22$ fixed, agreement with the true labels is moderate: ARI $= 0.66$, NMI $= 0.84$, homogeneity 0.82, completeness 0.85 and silhouette 0.35. Twelve crops keep at least 94% of their samples in one cluster (e.g. banana, coffee, mung bean, kidney beans), but several of those clusters also absorb a neighbouring crop: apple with grapes, and watermelon with muskmelon. Grapes, moth beans, pigeon peas and orange are split, with less than half their samples in their main cluster. Rice shares a cluster with jute, and the pulses (lentil, moth beans, pigeon peas, black gram) mix together (Appendix, Fig. 14).

5.4 Robustness

Feature importance and ablation. Humidity is the most important feature for every model, and rainfall is in every model's top three (Appendix, Fig. 15). Beyond these, the models rely on different nutrients: N for Random Forest, P for the Decision Tree and K for GNB. Temperature and pH matter little individually. Removing either costs at most 1.1 percentage points, and removing temperature costs Random Forest only 0.2 points. Rainfall is the feature no model can do without: retraining without it costs 2.5 points for Random Forest, 3.6 for GNB and 5.0 for KNN. Random Forest has the highest or joint-highest accuracy in every ablation.

Measurement noise. Table 4 and Fig. 5 show that the clean-data ranking does not hold under noise. Up to 10% noise all models except the single tree stay above 96%. At 20% noise, KNN keeps 94.7%, Random Forest and GNB fall to about 88%, and the Decision Tree drops to 77.8%. At 50% noise KNN still reaches 73.9%, against 55.6% for Random Forest and 50.5% for GNB.

Table 4: Test accuracy under Gaussian measurement noise (SD as a fraction of each feature's SD; mean of 20 repeats).

Noise	0	0.05	0.10	0.20	0.30	0.50
KNN	0.982	0.979	0.973	**0.947**	**0.894**	**0.739**
Decision Tree	0.980	0.956	0.913	0.778	0.650	0.455
Random Forest	**0.995**	0.986	0.965	0.878	0.761	0.556
Gaussian NB	**0.995**	**0.991**	**0.977**	0.882	0.741	0.505

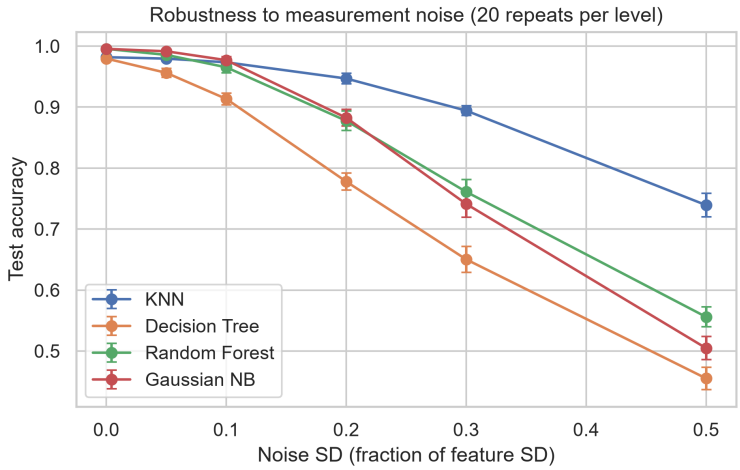


Figure 5: Accuracy as measurement noise increases. KNN degrades most gracefully; the single tree degrades fastest.

Data scarcity. With only three training samples per crop (66 in total), Random Forest reaches 93.0%, KNN 85.8%, GNB 79.6% and the Decision Tree 75.5% (Table 5). GNB overtakes Random Forest from about ten samples per crop. From 20 per crop onwards, Random Forest and GNB stay within 0.5 points of each other.

Probability quality. Table 6 and Fig. 6 show that Gaussian NB has the lowest log-loss, Brier score and top-label ECE in both evaluated conditions. This does not mean its errors are harmless: 94% of its clean predictions exceed 0.99 confidence, and its incorrect predictions average 0.79 confidence on clean data

Table 5: Test accuracy by number of training samples per crop (mean of 10 random subsets).

Per crop	3	5	10	20	40	80
KNN	0.858	0.890	0.932	0.950	0.974	0.982
Decision Tree	0.755	0.875	0.919	0.947	0.970	0.980
Random Forest	**0.930**	**0.966**	0.978	0.987	0.989	0.993
Gaussian NB	0.796	0.941	**0.985**	**0.989**	**0.993**	**0.995**

and 0.89 under noise. Random Forest is less confident on its errors (0.62 clean; 0.47 noisy), but its aggregate Brier score and ECE are worse. These metrics capture different aspects of probability quality, and the test-set results do not establish field calibration.

Table 6: Probability quality on clean and noisy (20%) test inputs. Brier is the mean summed one-hot squared error (0–2); ECE is ten-bin top-label expected calibration error; "Conf. err." is mean confidence on misclassified samples.

	Clean					20% noise				
Model	Log-loss	Brier	ECE	Conf. err.	>0.99	Log-loss	Brier	ECE	Conf. err.	>0.99
KNN	0.134	0.034	0.016	0.68	87%	0.42	0.079	0.020	0.74	82%
Decision Tree	0.737	0.041	0.021	1.00	100%	6.96	0.386	0.193	0.99	100%
Random Forest	0.079	0.023	0.062	0.62	24%	0.50	0.231	0.185	0.47	5%
Gaussian NB	**0.016**	**0.010**	**0.006**	0.79	94%	1.01	**0.203**	**0.089**	0.89	83%

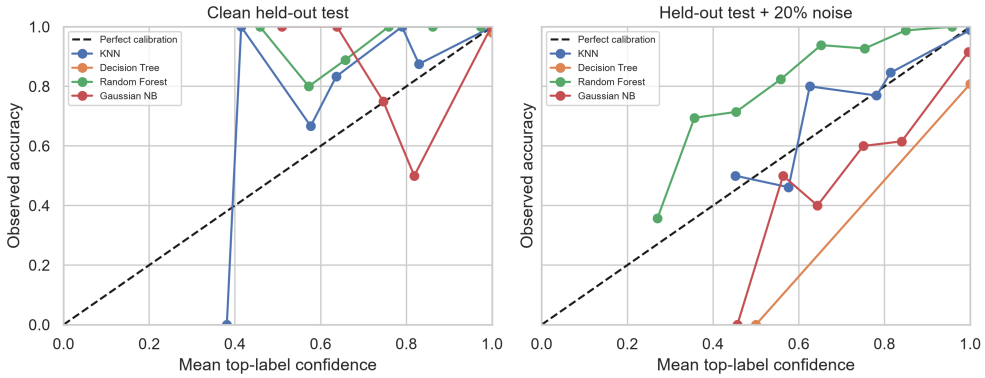


Figure 6: Top-label reliability diagrams on the clean test set and one fixed 20% noise realization. Points closer to the diagonal have confidence closer to observed accuracy; sparse bins reflect the small test sample and concentrated high-confidence predictions.

5.5 Prototype

Random Forest was retrained on all 2,200 samples and wrapped in a `recommend()` function. The function returns the three most probable crops and issues a warning for any input outside the range seen in training. An interactive form in the notebook lets a user set the seven values with sliders. For typical rice conditions the tool returns rice (0.87) followed by jute (0.12). For borderline conditions (rainfall 180 mm, humidity 80%) it returns jute (0.84) and rice (0.15), making the uncertainty visible to the user. For an out-of-range temperature of 48 °C it raises a warning and spreads its probabilities (papaya 0.28, mango 0.27, pigeon peas 0.16). Gaussian NB, by contrast, returned mango with probability 1.00 for the same input.

6 Discussion

6.1 Answering the research questions

RQ1. Soil and climate measurements identify the right crop among 22 with about 99.5% accuracy on this dataset, with nearly all remaining errors among agronomically similar crops. This matches the very high accuracies reported for similar data [6]. Section 6.3 argues, however, that the figure mainly reflects how the data was generated.

RQ2. No single technique is best on every criterion, so "best" depends on deployment conditions. On clean data, Random Forest and Gaussian NB are tied and significantly better than KNN and a single tree. With very little labelled data, Random Forest is best in the scarcity stress test; under heavy measurement noise, KNN is best. Probability scores disagree: GNB has lower aggregate log-loss, Brier score and ECE, while Random Forest is less confident on its errors. Neither model's probabilities should be treated as field-calibrated. Random Forest is retained for the prototype because its clean-data accuracy is tied for best and its tree ensemble performs well in the low-data stress test, while the probability display is explicitly advisory. These test-set stress-test results are descriptive and require confirmation on independent field data.

RQ3. The crops do not form 22 natural clusters. The strongest unsupervised structure is a two-way split, and K-means with $k = 22$ only partly recovers the labels (ARI 0.66). The partial agreement is agronomically meaningful, however. The groups K-means merges or splits—the pulses, rice and jute, the citrus and fruit crops, watermelon and muskmelon—are the same groups the classifiers confuse. Rice and jute are both warm, wet-season crops of the Indo-Gangetic plains with overlapping nitrogen and rainfall needs. Lentil, moth beans and black gram are low-nitrogen legumes with similar P, K and temperature ranges. PCA makes the same point: the space is *not* well separated in its main directions of variance, and classes become separable only when all seven dimensions are used. This explains why dimensionality reduction hurts and why a local method like KNN still performs well.

6.2 Why the models behave as they do

Gaussian NB performs far better than its violated assumption would suggest. Each crop occupies a compact box in feature space, and its fitted Gaussian is centred in that box. As long as the class with the largest likelihood is the correct one, classification succeeds, even though the densities are the wrong shape. This is the effect explained by Domingos and Pazzani [15]: Naive Bayes can be optimal under zero-one loss when its probability estimates are poor. The same mismatch explains the over-confidence. A product of seven Gaussian likelihoods falls off extremely quickly, so the posterior of the nearest class swamps all others. Naive Bayes is known to push probabilities towards 0 and 1 [26], and on this data the effect is extreme.

The single Decision Tree is limited by its axis-aligned, all-or-nothing splits. One misplaced threshold between overlapping pulse ranges misroutes every sample near the boundary, which explains both its pulse errors and its collapse under noise. Its leaves are nearly pure, so it reports full certainty even when wrong. *Random Forest* keeps the threshold-based representation but averages 100 decorrelated trees [14].

Its vote shares are less extreme on incorrect predictions than GNB's, but the aggregate Brier score and ECE do not favor the forest here; calibration is metric-dependent. Bootstrapping and random feature subsets also explain its strength with tiny training sets: each tree sees a different view of the few samples available, while GNB cannot estimate per-class variances reliably from three points.

KNN with distance weighting and Manhattan distance is the most noise-tolerant model. Its prediction depends on the local neighbourhood rather than on a sharp boundary. A noisy query usually remains closer to samples of its own crop than to those of others, because the box-shaped classes are much larger than the 20% noise shift. Its clean-data accuracy is lower because, near the boundaries between overlapping crops, five neighbours can be outvoted by another class.

6.3 Dataset quality and what the accuracy really means

The evidence that the dataset is synthetic is the most important finding for interpreting this and other work that uses it. Uniform within-class distributions with hard limits make the task close to looking up a crop's published requirement ranges. The 99%+ accuracies reported for this dataset, including those in this report, should therefore be read as an upper bound under ideal conditions, not as expected field performance. Real soil tests have laboratory error. Weather values are seasonal averages that vary from year to year. Real farms often grow crops outside textbook ranges because of irrigation, variety selection or market pressure. All of these would make classes overlap and blur the boundaries.

The robustness experiments are a first step towards quantifying that gap. With noise of only 20% of a feature's SD (about 11 mm of rainfall or 0.15 pH units), the best clean-data models lose about 11 percentage points. Ranking models by clean accuracy alone would have chosen GNB, which degrades more steeply and stays confident when it is wrong. This is the main methodological lesson for researchers working with this or similar benchmark datasets: when headline accuracy is saturated, models should be compared on robustness, data efficiency and calibration.

6.4 Comparison with prior work

The clean-data accuracy reported here is in line with the high figures in the literature. Kulkarni et al. [6] reported above 99% with a majority-voting ensemble, and earlier systems on regional data, such as those of Pudumalar et al. [5] and Rajak et al. [8], also favoured ensembles of heterogeneous classifiers. The present results support part of that preference. Random Forest, itself an ensemble, is strong, and a single tree is clearly weaker than an ensemble of trees. They also qualify it: Gaussian Naive Bayes matches the ensemble on clean inputs, so high accuracy alone does not show that an elaborate model is needed. Unlike the studies cited, this report gives variance over 50 folds and significance tests; these show that the small differences between the best models are not meaningful. It also probes noisy inputs and scarce training data. The probability results demonstrate that confidence scores should not be presented as calibrated probabilities without further validation: proper scores, top-label ECE and confidence on errors give different views of model behavior. A ranked shortlist can expose alternatives, but its scores need clear caveats and validation on independent field data. Doshi et al. [7] additionally argue that recommendations should consider factors beyond soil and weather, such as market prices.

6.5 Significance and applications

The study is not a "toy" exercise. Crop choice affects household income, food security and fertiliser use for millions of smallholders, and decision-support tools of this kind are being built and deployed [2, 7]. The study improves on prior work in three practical ways. First, it gives a reproducible protocol (leakage-free pipelines, repeated CV and corrected tests) that other researchers can reuse to compare algorithms fairly. Second, it shows that headline accuracy on the popular public dataset for this task is essentially saturated and likely reflects how the data was generated. Third, it shows why robustness and probability evaluation matter before deployment. For agricultural extension services and app developers, the results suggest several practices: present a ranked shortlist with clear warnings that scores are uncalibrated; prioritise reliable rainfall and humidity data, which matter most, over less informative inputs such as pH; and validate on local field data before trusting any benchmark figure.

6.6 Limitations

The main limitation is the dataset itself: it appears synthetic, has no geographic or temporal information, and covers only 22 crops, mostly suited to Indian conditions. The noise experiment adds independent Gaussian noise to the test inputs only. Real measurement errors may be correlated, biased or heavy-tailed, and models could be trained with noise to improve robustness. Only four classifiers were compared, under modest grid-search budgets. Support vector machines, gradient boosting and calibrated variants of each model were not included. The "most suitable crop" label ignores economic factors such as market prices and input costs, which Doshi et al. [7] show are important to farmers. Finally, no real farmers or agronomists evaluated the prototype.

7 Social, Ethical, Legal and Professional Considerations

Social impact and access. A recommender could help smallholders who lack access to agronomists, but only if they can supply its inputs. Soil N, P, K and pH require laboratory or kit-based testing that costs money and time. If only better-off farmers can afford this, the tool may widen the gap it aims to close. Deployment through government soil-testing schemes or cooperative extension services would reduce this risk. The interface must also work in local languages and on basic phones.

Harm from wrong advice. An incorrect recommendation can cost a farmer a season's income. The prototype therefore shows a ranked shortlist and warns about inputs outside the training range. If probability scores are displayed, they must be labelled as uncalibrated model outputs: the models can be confident when wrong, and calibration metrics disagree (Table 6, Fig. 6). Recommendations should support, not replace, the judgement of farmers and local agronomists, and the tool should state clearly that its advice must be checked against local knowledge.

Data provenance and representativeness. The training data appears synthetic and its geographic origin is undocumented. Using it to advise real farmers, especially outside the region its ranges were drawn from, would be misleading. The data's origin and limitations should be disclosed to users. The licence terms on the Kaggle page should be checked before any commercial use, and the source must be cited, as it is

here [10].

Privacy and legal issues. A deployed system would collect field locations, soil profiles and possibly farmers' identities and yields. Such data is commercially valuable and could be misused, for example in land or credit decisions. Collection should be minimal and consent-based. In Nepal it should comply with the Individual Privacy Act, 2075 (2018), and with equivalent data-protection law elsewhere. Liability for losses caused by faulty advice should be stated clearly in the terms of use.

Environmental considerations. Recommendations that ignore sustainability could encourage nutrient-depleting rotations or water-intensive crops in drought-prone areas. A responsible system should consider soil health and water availability, not only immediate suitability.

Professional conduct. The work follows professional codes of conduct such as that of the BCS, the Chartered Institute for IT, in reporting results honestly, including negative and limiting findings. Examples are the evidence that the data is synthetic and the loss of accuracy under noise. All code is shared for reproducibility, and all sources are acknowledged.

8 Conclusion and Future Work

This report compared KNN, Decision Tree, Random Forest and Gaussian Naive Bayes for recommending one of 22 crops from soil and climate data, under a leakage-free tuning protocol with repeated cross-validation and significance testing. Random Forest and Gaussian NB reach 99.55% test accuracy and are statistically tied. PCA and K-means show that the remaining errors lie among agronomically similar crops, such as rice and jute and the pulses, that also overlap in unsupervised structure. Exploratory analysis suggests the dataset is synthetic, so its near-perfect accuracies should not be read as field performance. Stress tests show that clean-data rankings do not transfer automatically: KNN is most tolerant of noise and Random Forest performs best with few training samples. Probability metrics disagree: GNB scores better on aggregate log-loss, Brier and ECE, yet is more confident on its errors. These post-hoc test-set diagnostics are descriptive, and no model is established as calibrated for field use. Random Forest is used in a prototype that returns a ranked shortlist and warns on out-of-range inputs.

Future work should validate the approach on real field data with geographic and seasonal information, such as Nepali soil-testing records. Other directions are training with realistic noise models, adding probability calibration (e.g. isotonic regression) and stronger learners such as gradient boosting, and including economic and sustainability factors in the recommendation. A user study with farmers and extension officers would show whether a probability-annotated shortlist actually improves decisions.

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A Appendix: Evidence of Running the Experiments

All results above come from the notebook `Raju_Shah_260026_CropRecommendation.ipynb`, submitted with this report together with run instructions (`README.md`). The notebook executes end to end without errors and stores all outputs. This appendix reproduces the figures that are not shown in the main text.

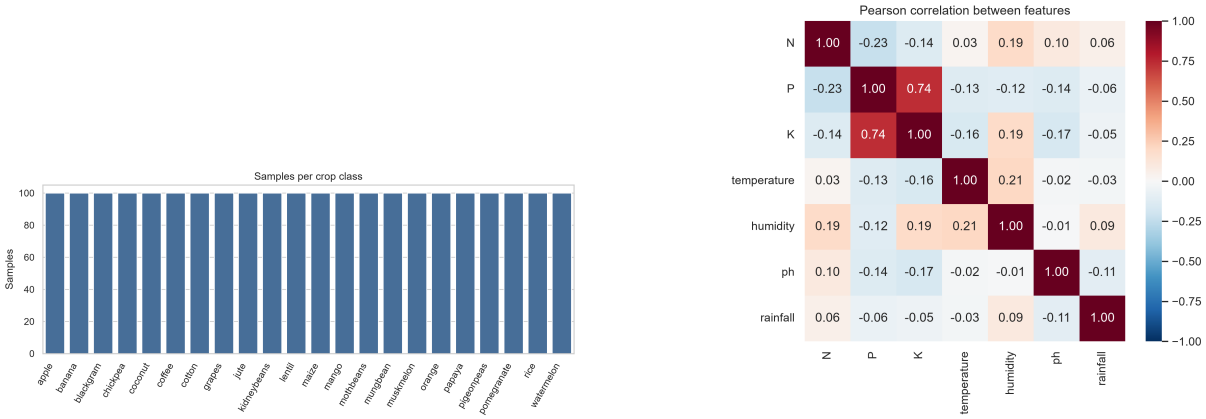


Figure 7: Left: class balance (100 samples per crop). Right: feature correlations; only P–K is strongly correlated.

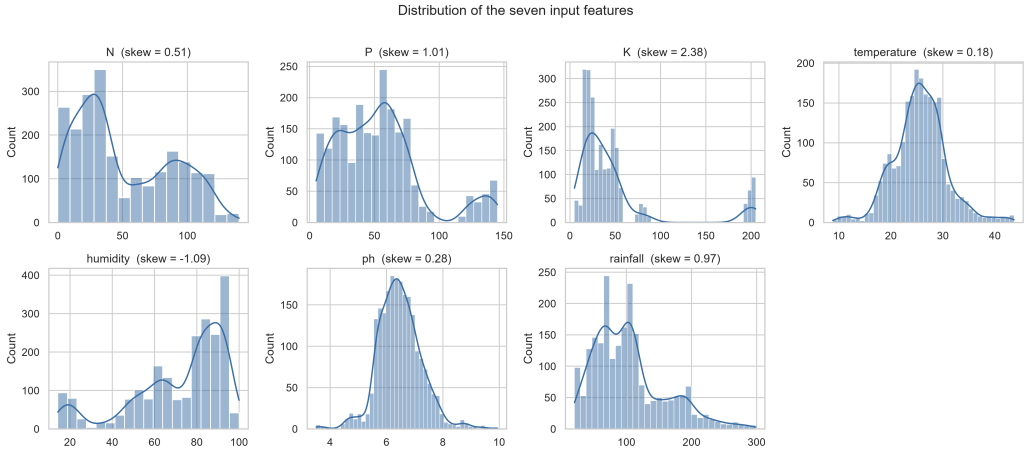


Figure 8: Distributions of the seven features over the whole dataset.

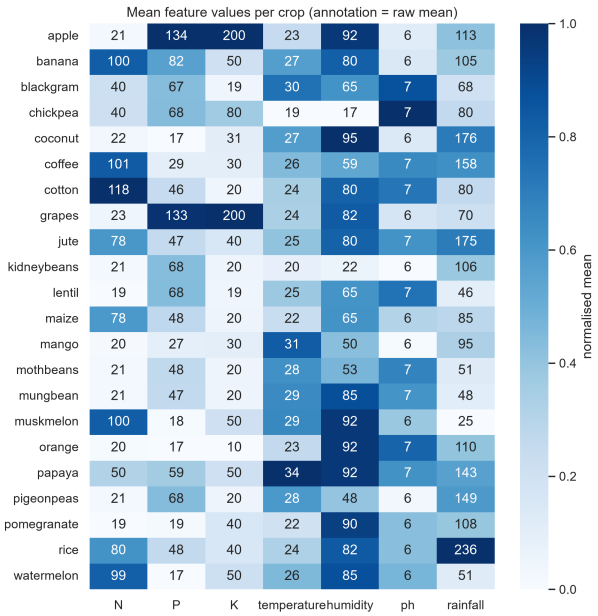


Figure 9: Mean feature values per crop (colour normalised per feature; annotations show raw means).

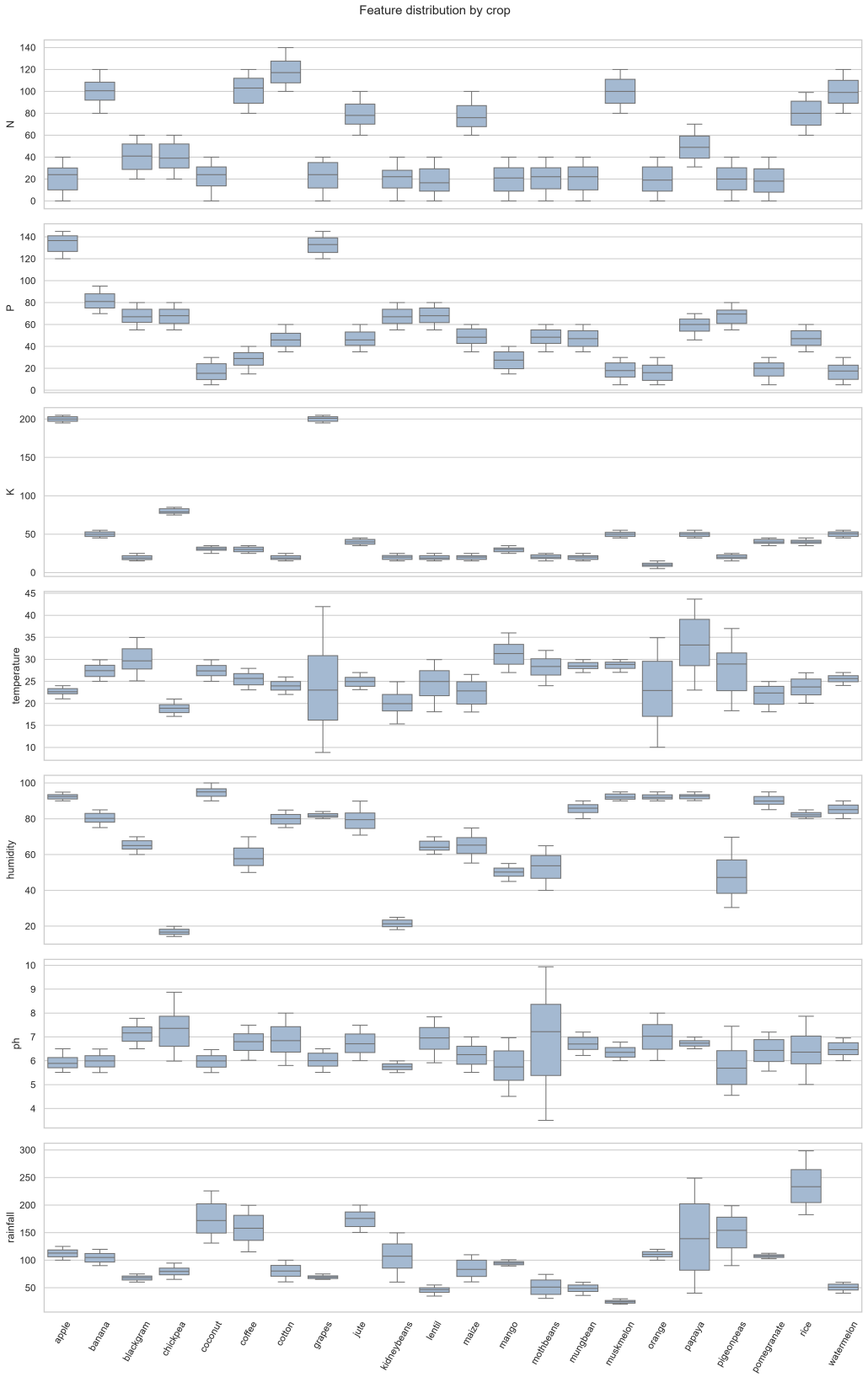


Figure 10: Feature distributions by crop.

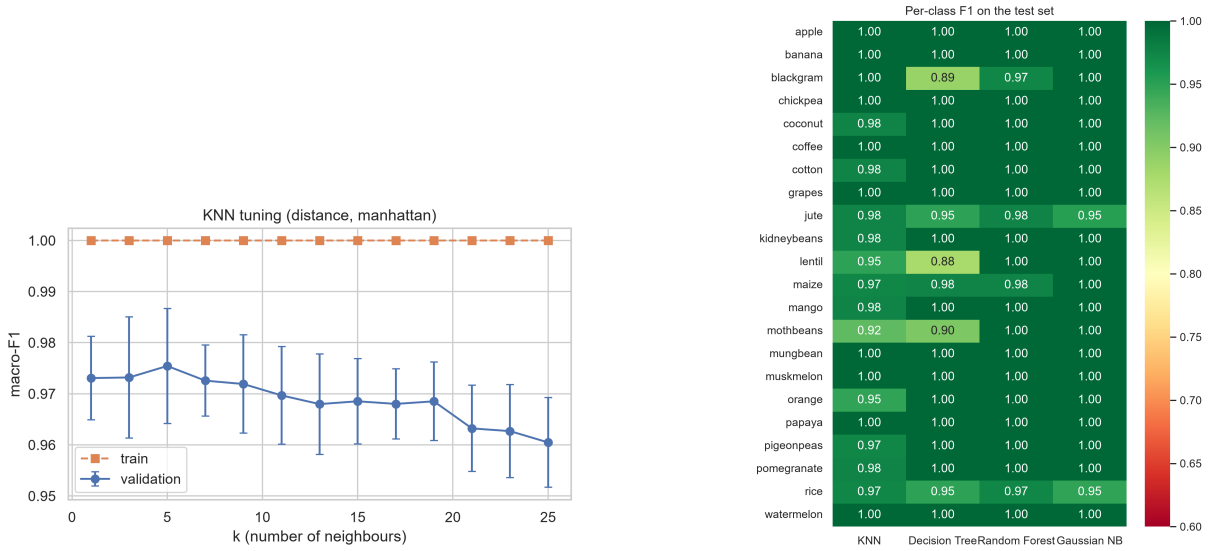


Figure 11: Left: KNN validation and training macro-F1 against k (distance weights, Manhattan metric). Right: per-class test F1 for each model.

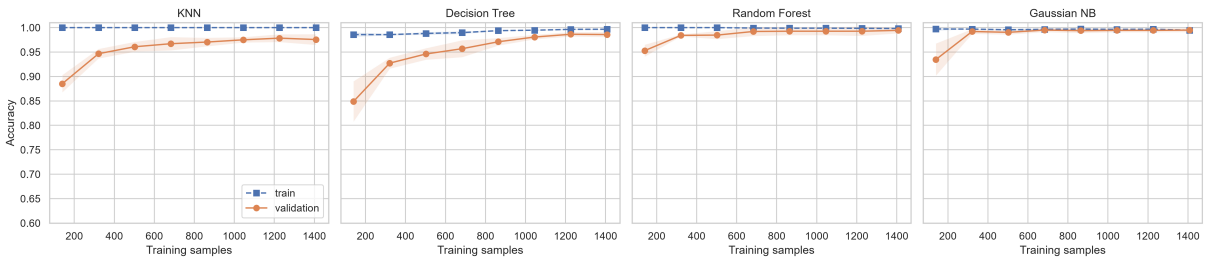


Figure 12: Learning curves (5-fold CV on the training set).

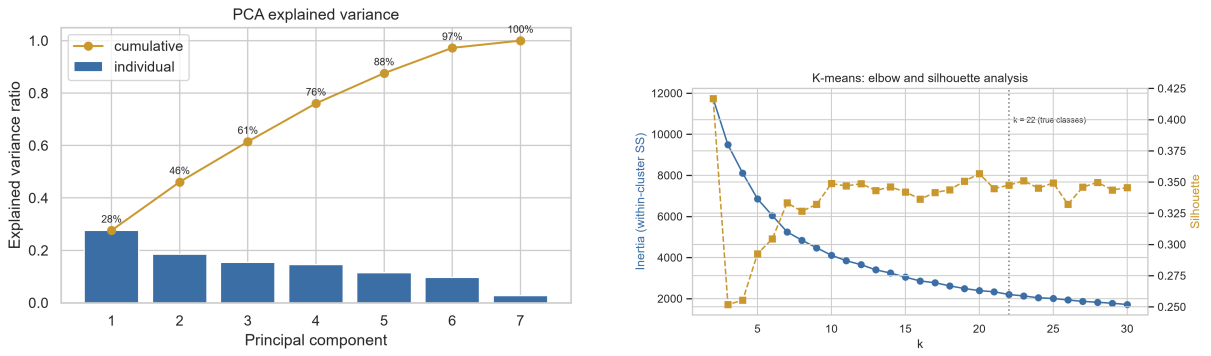


Figure 13: Left: PCA explained variance. Right: K-means inertia and silhouette for $k = 2$–30.

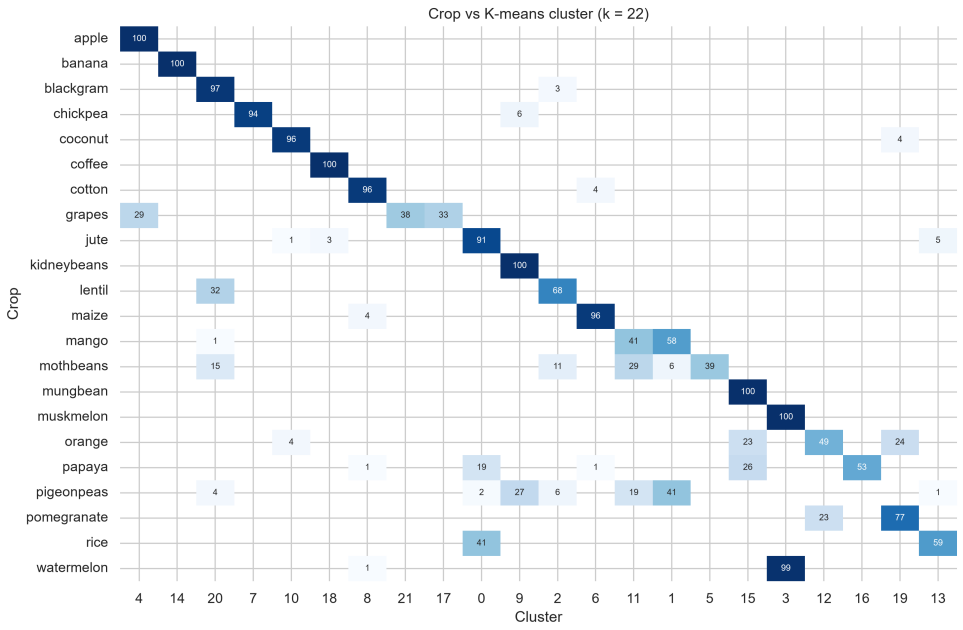


Figure 14: Crop vs. K-means cluster ($k = 22$). Clusters are ordered by dominant crop.

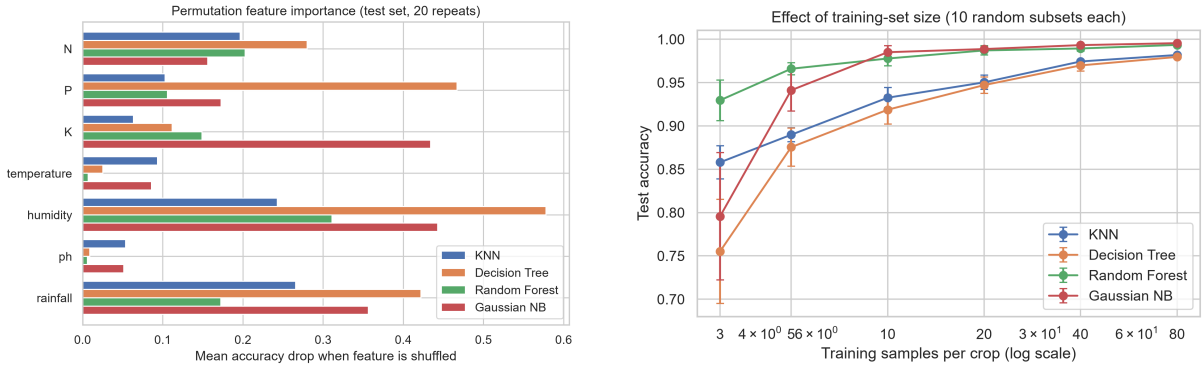


Figure 15: Left: permutation feature importance on the test set. Right: accuracy by number of training samples per crop.

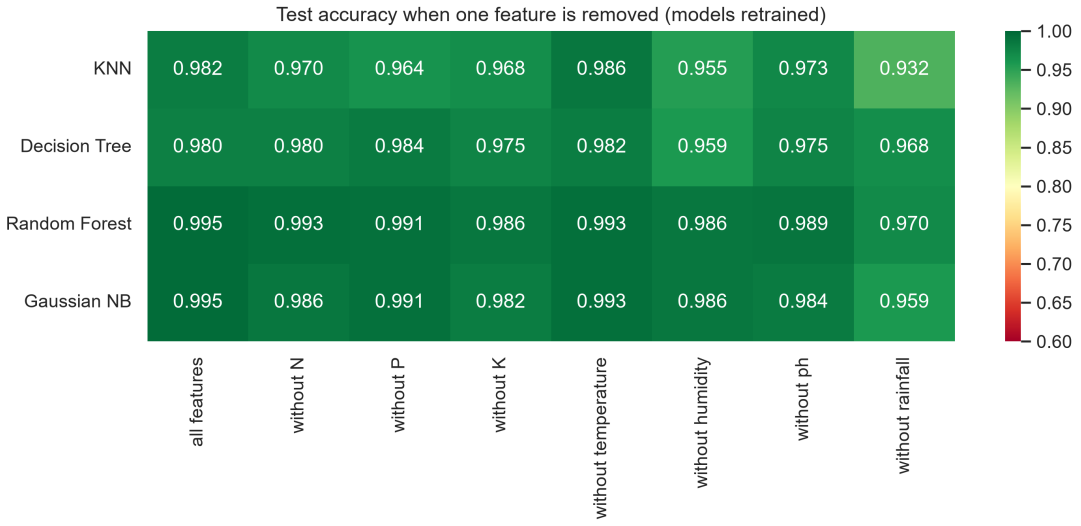


Figure 16: Test accuracy when each feature is removed and the model retrained.